EXPERIMENT 4A

Information Retrieval System using TF-IDF and Cosine Similarity

Learning Objectives :

The objective of this experiment is to develop an Information Retrieval system using Natural Language Processing techniques and the NLTK library. The experiment helps in understanding text pre-processing operations such as lowercasing, punctuation removal, tokenization, and stop-word removal. It also helps in converting textual documents into numerical representations using TF-IDF and retrieving the most relevant reviews using cosine similarity.

Methods Used / Python Libraries Involved and Purpose :

Pandas is used to load and process the Amazon Fine Food Reviews dataset and to access the review text.

NLTK is used for Natural Language Processing operations such as tokenization and stop-word removal.

String is used to remove punctuation from the review text during pre-processing.

TfidfVectorizer is used to convert the cleaned review documents into numerical TF-IDF vectors.

Cosine Similarity is used to measure the similarity between the query and the review documents.

Text Pre-processing is performed by converting text to lowercase, removing punctuation, tokenizing the text, and removing stop-words.

TF-IDF Vectorization represents the cleaned textual reviews numerically based on the importance of words in the documents.

Cosine Similarity compares the query vector with the review vectors and identifies the reviews that are most similar to the query.

PROGRAM CODE :

```
import pandas as pd
import string
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('punkt_tab')
```

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```
!wget -q https://www.kaggle.com/api/v1/datasets/download/snap  amazon-fine-food-reviews -O
reviews.zip
!unzip -o -q reviews.zip
df = pd.read_csv("Reviews.csv")
reviews = df['Text'].dropna().head(5000)
stop_words = set(stopwords.words('english'))
def preprocess(text):
    text = text.lower()
    text = text.translate(str.maketrans('', '', string.punctuation))
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)
cleaned_reviews = reviews.apply(preprocess)
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(cleaned_reviews)
query = reviews.iloc[0]
cleaned_query = preprocess(query)
query_vector = vectorizer.transform([cleaned_query])
similarity_scores = cosine_similarity(
    query_vector,
    tfidf_matrix
).flatten()
top_k = 5
top_indices = similarity_scores.argsort()[-top_k:][::-1]
print("\nTop Relevant Reviews:\n")
for i, index in enumerate(top_indices):
    print("Rank:", i + 1)
    print("Similarity Score:", round(similarity_scores[index], 4))
    print("Review:", reviews.iloc[index])
    print("-" * 80)
```

OUTPUT :

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]  Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]  Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]  Unzipping tokenizers/punkt_tab.zip.
```

Top Relevant Reviews:

Rank: 1
Similarity Score: 1.0
Review: I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labrador is finicky and she appreciates this product better than most.

Rank: 2
Similarity Score: 0.3269
Review: This canned food has real chunks of chicken in it. Dogs love it. Looks and smells better than human-quality canned beef stew (like Dinty Moore).

Rank: 3
Similarity Score: 0.1958
Review: I have purchased this product once before from Amazon. It was delivered promptly and in tact. This is the only food our dog will eat and we used to purchase it at PetSmart in New York but in Arizona, it is no where to be found. This food is holistic and has no unnatural products at all. We try to give our dog good quality dog food and this was it. So now we go through Amazon and our dog is quite happy. Bought a double amount to receive free shipping. It was a very good deal.

Rank: 4
Similarity Score: 0.1696
Review: I like this coffee. It smells good while brewing and tastes as good as it smells. for a decaf coffee it is very good
will definitely buy this at my local store.

Rank: 5
Similarity Score: 0.1651
Review: My goldendoodle is picky and likes to change up his food once in a while. So I decided to try two other brands....what a mistake.
Newman's own is really the best: my dog looks better, feels better, and the quality of food is very good. Do not do like I did and experiment around...my poor dogs stomach....stick with Newmans.

Learning Outcome :

From this experiment, I learned how to develop an Information Retrieval system using NLP techniques and NLTK. I gained practical knowledge in text pre-processing, TF-IDF vectorization, and cosine similarity to retrieve and analyze the most relevant reviews based on similarity scores. I also learned how to obtain the Amazon Fine Food Reviews dataset directly from Kaggle and use the retrieved similarity scores to rank the relevant reviews.